Secure College Campus Network Design and Implementation Using Cisco Packet Tracer

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***Abstract*—This paper presents the comprehensive design, configuration, and functional validation of a small-scale college campus network utilizing the Cisco Packet Tracer simulation environment [1]. The observed network topology incorporates a Cisco 2911 router, a central switch, and three distinct user Virtual LANs (VLANs) designated for Administration, Computer Science and Engineering (CSE), and Electronics and Communication Engineering (ECE). Furthermore, a dedicated server VLAN is deployed to host essential shared services, including DHCP, DNS, HTTP, and email operations. Network screenshots reveal that the user networks are systematically labeled as 192.168.10.0/24, 192.168.20.0/24, and 192.168.30.0/24, while the server network operates under the label 192.168.70.0/24. The documented evidence includes a `trunk-status` command verifying that VLANs 10, 20, 30, and 70 are allowed and active on interface GigabitEthernet0/1. Additionally, testing evidence demonstrates a successful ICMP ping to the 192.168.10.1 gateway, a ping to 192.168.70.3 resulting in three successful replies and one initial timeout, and a client web browser successfully rendering the college website. Analysis of the DHCP screenshots also exposes configuration values that require immediate correction, specifically noting a default gateway and DNS server erroneously set to 0.0.0.0. Ultimately, this project demonstrates a highly practical, segmented campus-network prototype and establishes a repeatable blueprint for completing comprehensive service validation.**

***Index Terms*—Campus network, Cisco Packet Tracer, VLAN, IPv4 addressing, DHCP, DNS, HTTP, network security.**

I. INTRODUCTION

In modern educational institutions, college networks play a critical role in supporting a wide array of activities, including teaching, administrative operations, student computing, and the provisioning of shared network services [2]. A common architectural challenge arises when all network endpoints within such an environment are placed into a single broadcast domain. Under those unsegmented conditions, it becomes significantly harder for administrators to effectively separate departmental traffic, leading to potential operational inefficiencies. Furthermore, a single broadcast domain complicates the process of troubleshooting network faults and hinders the ability to apply different, granular access policies tailored to specific user groups.

To address these challenges, Virtual LANs (VLANs) are utilized to provide logical separation across a shared physical switching infrastructure, while standard routing mechanisms allow for controlled and secure communication between these disparate subnets. Alongside these foundational routing and switching operations, common application-level services—such as Dynamic Host Configuration Protocol (DHCP), Domain Name System (DNS), web hosting, and email—are necessary to support everyday campus operations.

This project specifically models a small campus infrastructure within Cisco Packet Tracer. The supplied topology screenshot provides evidence of one central switch connected to a single Cisco 2911 router, serving six interconnected PCs and four dedicated servers. The primary purpose of this paper is to thoroughly document the network design as it is shown in the provided evidence, describe the underlying configuration, identify functional issues that should be corrected, and explicitly outline the security controls capable of being validated. The scope of this paper is defined as presenting a functional educational prototype; it explicitly does not constitute a claim of achieving production-grade security or formally measured network performance.

II. RELATED WORK

Prior studies investigating campus-network architectures have frequently utilized network simulation to effectively illustrate VLAN-based segmentation, inter-VLAN routing, and essential service configuration [1]. For instance, Ahmed and Al-Hamadani actively described a secure campus network that was simulated heavily using Cisco Packet Tracer. Similarly, Varne et al. presented an extensive campus-network design and practical implementation utilizing Packet Tracer as the primary testing ground [2]. Furthermore, Fitria et al. discussed the meticulous construction of campus networks emphasizing VLAN deployments, detailed subnetting, DHCP allocation, and strict connectivity verification [3].

From a standards perspective, IEEE 802.1Q formally defines the mechanisms for VLAN tagging and bridged-network operations, which form the backbone of the logical separation in this project [4]. Additionally, IEEE 802.1X specifies port-based network access control protocols, a standard highly relevant to enterprise-scale deployments that possess suitable authentication infrastructures [5].

The current network project actively uses VLAN identifiers and relies on a configured trunk link. However, based strictly on the supplied evidence alone, the project does not establish that IEEE 802.1X authentication or every single recommended security control has been fully configured. The core practical contribution of this project remains its highly accessible demonstration of departmental subnetting protocols, the establishment of a dedicated server VLAN, and successful HTTP service access achieved via hostname resolution.

III. REQUIREMENTS AND METHODOLOGY

The underlying design of this prototype was carefully organized around four distinct user-facing requirements: (1) To physically and logically separate departmental endpoint groups utilizing VLANs; (2) To provide a highly dedicated server segment meant specifically for shared services; (3) To enable client machines to reliably reach their local gateway alongside permitted external services; and (4) To thoroughly document the network state through the extensive use of visual screenshots and CLI output captures.

The primary implementation environment chosen to execute these requirements is Cisco Packet Tracer. To fulfill these objectives, the research methodology was divided into six sequential stages. First, all necessary devices and logical groups were carefully placed onto the simulator workspace. Second, specific IPv4 subnets were directly associated with the Administration, CSE, ECE, and dedicated server groups. Third, all VLANs were actively configured on the infrastructure, and the relevant switch link was rigorously checked to verify proper trunking operations. Fourth, the central router, required gateway interfaces, and vital server services were configured. Fifth, detailed client-side tests were actively performed using both standard ICMP ping commands and a simulated web browser. Sixth, all resulting screenshots were critically reviewed to identify gaps between the intended design and the actual observed configuration.

The supplied visual screenshots provide concrete evidence supporting the topology, displaying the DHCP service configuration panel, showcasing an HTTP web page load, and proving baseline ICMP connectivity. Importantly, they do not include the complete running configuration, detailed Access Control List (ACL) output, or a Network Address Translation (NAT) table. These specific missing items are formally listed as further validation tasks rather than being falsely claimed as verified results.

IV. NETWORK ARCHITECTURE AND ADDRESSING

The highly visible topology of the network actively uses a singular central switch as the primary aggregation point for the infrastructure. A Cisco 2911 router is directly connected above this switch, while a cloud object is utilized to represent the broader external network. The switch itself connects directly to the various departmental PCs and the four essential service hosts.

The topology labels distinctly identify VLAN 10 for Administration, VLAN 20 for CSE, VLAN 30 for ECE, and VLAN 70 designated specifically for the servers. Correspondingly, the distinct network labels are listed as 192.168.10.0/24, 192.168.20.0/24, 192.168.30.0/24, and 192.168.70.0/24. A /24 subnet formally utilizes the 255.255.255.0 subnet mask and is capable of providing 254 conventional host addresses for device assignment. The captured ping evidence successfully confirms that the IP 192.168.10.1 actively responds, acting as a functional gateway interface.

TABLE I. OBSERVED NETWORK PLAN

Group/VLAN	Network shown	Purpose
VLAN 10	192.168.10.0/24	Administration endpoints
VLAN 20	192.168.20.0/24	CSE endpoints
VLAN 30	192.168.30.0/24	ECE endpoints
VLAN 70	192.168.70.0/24	DHCP, DNS, HTTP, and email
Trunk	Gi0/1	Carries VLANs 10, 20, 30, 70

A. VLAN and Trunk Observations

The detailed switch CLI screenshot officially reports that interface GigabitEthernet0/1 currently exists in a trunking state and actively utilizes standard 802.1Q encapsulation. The default native VLAN shown in the configuration is VLAN 1. Concurrently, VLANs 10, 20, 30, and 70 are explicitly listed as allowed, active, and functioning in the spanning-tree forwarding state on this specific trunk interface. This constitutes direct and undeniable evidence that the displayed trunk is actively carrying traffic for the four specified VLANs.

For the purposes of improved network hardening, the native VLAN should ideally be deliberately selected and ensured to be consistent at both terminating ends of the link. Furthermore, the allowed VLAN list should be explicitly restricted to include only those VLANs functionally needed on that specific link. If VLAN 1 is determined to not be required for standard user traffic, administrators should consider moving the native VLAN to a fully dedicated, unused VLAN to reduce avoidable security exposure.

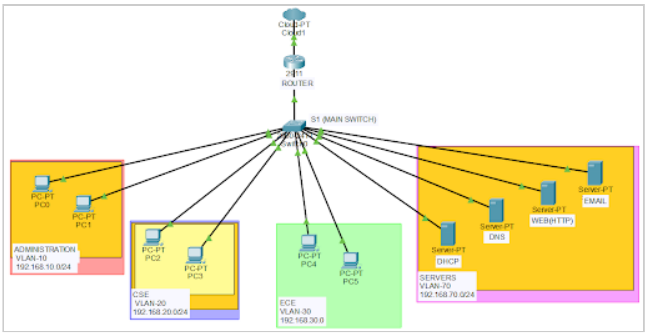


Fig. 1. Cisco Packet Tracer topology showing the router, central switch, departmental endpoint groups, and four servers.

V. IMPLEMENTATION

A. VLAN Segmentation and Switching

The logical device groups explicitly mapped within the topology are Administration, CSE, ECE, and Servers. Each unique group is fundamentally associated with a completely separate VLAN and an isolated IPv4 subnet. Accordingly, the end-device ports must be correctly configured as access ports resting within their intended target VLAN. Any uplink ports bridging switches, where they are present, are strictly required to utilize standard 802.1Q trunking alongside an explicit allowed-VLAN list. While the provided screenshot fully verifies the existence of one trunk on Gi0/1, it inherently fails to show every individual access-port assignment. As a result, those specific assignments must be manually checked utilizing standard `show vlan brief` administrative commands on the core switch.

```
Campus-Switch>show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      10,20,30,70

Port        Vlans allowed and active in management domain
Gig0/1      10,20,30,70

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      10,20,30,70

Campus-Switch>
```

Fig. 2. Trunk status for GigabitEthernet0/1 showing 802.1Q trunking and VLANs 10, 20, 30, and 70 allowed and active.

B. IP Addressing and Routing

The visibly displayed gateway connectivity test directed to the 192.168.10.1 address succeeds without issue. To ensure comprehensive inter-VLAN communication, the overall topology must feature a functional and verified Layer 3 routing design. This can be achieved through either a router-on-a-stick model leveraging router subinterfaces and a trunk leading back to the router, or by utilizing a multilayer switch outfitted with Switched Virtual Interfaces (SVIs) and explicitly enabled IP routing. The supplied singular image does not automatically reveal which precise routing method is utilized in this architecture, nor does it inherently prove that all distinct VLANs possess the ability to communicate with one another.

C. DHCP Configuration Analysis

An analysis of the Server DHCP panel visibly shows one pool distinctly named "ADMIN" and an entirely separate pool named "serverPool". The actively selected serverPool explicitly displays a starting IP address of 192.168.70.0, a standard subnet mask of 255.255.255.0, a default gateway set to 0.0.0.0, and a DNS server simultaneously set to 0.0.0.0.

These specific, displayed values absolutely should not be presented as representing a functionally correct client pool. The specific address ending in .0 acts as the designated network address within a standard /24 subnet and fundamentally should not be dynamically assigned to an individual host. Similarly, the placeholder address 0.0.0.0 is entirely invalid as an intended client gateway or functioning DNS server for ordinary network clients.

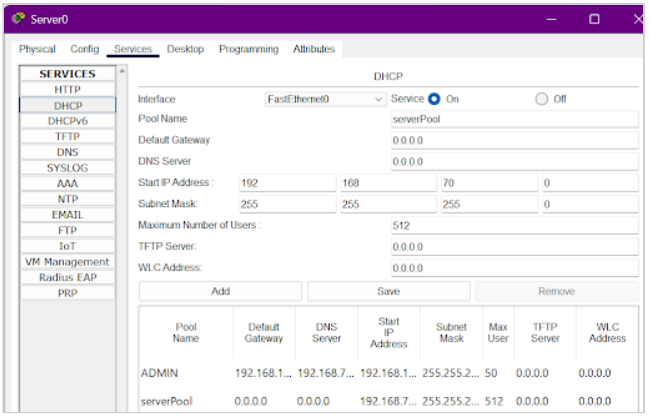


Fig. 3. DHCP service configuration screen. The displayed serverPool values require correction or confirmation before the pool is used.

To rectify this, administrators must actively configure the server pool utilizing a perfectly valid host start address, insert the actual valid gateway address meant for VLAN 70, and apply the true IP address of the acting DNS server. Any statically assigned server IP addresses must be manually excluded from these dynamic assignment ranges. For the various departmental VLANs, it is required to create an entirely separate scope for each subnet.

D. DNS and HTTP Services

The provided web browser screenshot successfully shows the requested address http://www.college.com and effectively renders a page prominently headed "College Campus Network," alongside the supplementary text "Welcome to the college website." This specific visual serves as hard evidence that a functioning web page was successfully displayed utilizing the hostname inside the Packet Tracer simulated browser.

However, to ensure this test is entirely reproducible, it is necessary to rigorously verify that the operational DNS service contains an accurate record for www.college.com that correctly points back to the web server's actual, static IP address. For stronger validation evidence, a direct screenshot of the internal DNS record should be preserved rather than assuming the lookup succeeded flawlessly.

E. Email Service Implementation

A dedicated, isolated server is visibly labeled as "EMAIL" within the overarching topology. To definitively validate the operational status of email, an administrator must actively enable the appropriate application service, accurately configure both the parent domain and the individual user accounts, configure all client-side incoming and outgoing server connection details, and ultimately send a test message securely between two distinct test accounts to prove end-to-end functionality.

F. Router and External Connectivity

The primary routing device is shown directly connected to a generalized cloud object. Despite this visual, the comprehensively supplied screenshots unfortunately do not display its configured interface addresses, its active routing table, any potential NAT/PAT configurations, or a single successful external-network transmission test. Under no circumstances should external connectivity or complete NAT success be claimed without utilizing a dedicated test host.

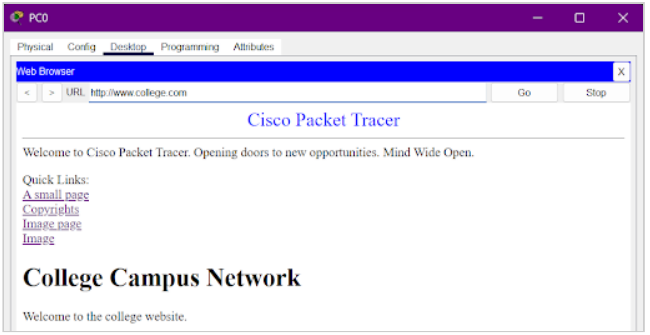


Fig. 4. HTTP test from a client browser showing the college campus web page at www.college.com.

VI. TESTING AND RESULTS

The technical results outlined below are strictly limited by what can be directly read and interpreted from the submitted screenshots. The first initial ping command presented in the screenshot is factually malformed, predictably returning an "Invalid Command" error. The immediately subsequent command, ping 192.168.10.1, is formally valid and successfully receives exactly four replies.

Following this, a secondary connectivity test directed to 192.168.70.3 notably receives one initial timeout prior to receiving three successful replies. This precise outcome yields a total of 3 successfully received replies out of 4 overall attempts, directly resulting in a 25% observed packet loss rate during that single specific run.

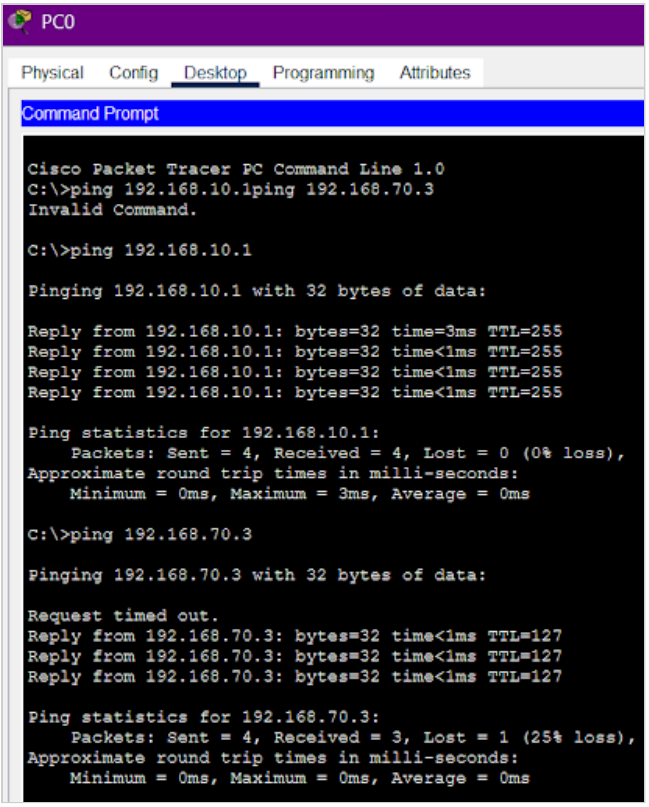


Fig. 5. ICMP connectivity tests: successful ping to 192.168.10.1 and a test to 192.168.70.3 with one timeout and three replies.

TABLE II. VALIDATION RESULTS BASED ON PROVIDED EVIDENCE

Test	Evidence observed	Status/Interpretation
Topology	Central switch, router, 6 PCs, 4 servers	Observed
Trunk status	Gi0/1 trunking; 802.1Q: VLANs active	Pass for displayed trunk
Gateway ping	192.168.10.1: 4 sent, 0% loss	Pass for this test
Server ping	192.168.70.3: 4 sent, 3 received, 1 lost	Partial; repeat test
HTTP page	college.com displays college page	Page displayed; capture DNS
DHCP pool	serverPool shows gateway/DNS 0.0.0.0	Needs correction
Email check	No send/receive evidence supplied	Not verified
ACL/SSH/NAT	No relevant evidence supplied	Not verified

A. Result Interpretation

The provided evidentiary materials strongly support the existence of basic baseline connectivity leading to the 192.168.10.1 interface, successfully active VLAN carriage operating on the visibly displayed trunk, and highly successful HTTP page rendering triggered by hostname resolution.

As noted, the very first ICMP response originating from 192.168.70.3 officially times out, while the subsequent three consecutive replies manage to succeed. This initial ping timeout can frequently occur due to active address resolution processes (ARP) or standard first-packet setup delays occurring within a simulator. This highly specific behavior must be described accurately within the documentation and should absolutely be re-tested for consistency.

The visually provided DHCP configuration screen stands as the single most visible configuration concern in the current network state. A fully comprehensive network validation should meticulously include a specialized test pair for every single configured access policy. Specifically, there must be one flow that is mathematically expected to succeed, and an opposing flow that is expected to be explicitly blocked. Executing a ping alone is entirely insufficient to adequately validate complex application services or a rigorous security policy.

VII. SECURITY CONSIDERATIONS

The implementation of rigorous VLAN segmentation inherently reduces the administrative need to place every single network user within one massive broadcast domain. However, the presence of VLANs alone does not natively enforce a fully complete or holistic security policy. Standard inter-VLAN traffic absolutely should be tightly controlled utilizing Access Control Lists (ACLs) crafted precisely according to the core principle of least privilege.

As a primary example, standard student endpoints logically may be granted access to approved web interfaces, DNS resolution, and specific academic learning services, while being simultaneously and explicitly denied access to critical administrative endpoints and network device-management interfaces. Crucially, they should be definitively claimed as formally implemented only after the relevant ACL configurations, and both the permitted and denied validation tests, have been fully executed and successfully captured.

For device administration, all management access must strictly utilize Secure Shell (SSH) in place of the inherently insecure Telnet protocol. Broad switch hardware hardening techniques can additionally include manually disabling all entirely unused ports, aggressively assigning these unused ports directly to an isolated non-user VLAN, heavily configuring dedicated port security protocols upon the active edge ports, and proactively enforcing spanning-tree protections such as applying BPDU Guard on edge connections.

Presently, the supplied current topology screenshot completely fails to visually show a highly separated management VLAN, a dedicated unused VLAN, or any evidence of a comprehensive ACL policy. Therefore, these specific advanced controls operate solely as proposed conceptual improvements rather than being accepted as verified, actionable project results.

VIII. DISCUSSION AND LIMITATIONS

This foundational network design successfully illustrates a straightforward, highly logical methodology for efficiently organizing a small college campus network precisely around department-based VLANs and an entirely dedicated, separated service segment. The captured trunk CLI output serves as highly useful physical evidence effectively proving that VLANs 10, 20, 30, and 70 are technically allowed and functionally active simultaneously on the Gi0/1 interface. Additionally, the highly successful gateway ping clearly demonstrates confirmed reachability, and the captured browser screenshot functionally demonstrates that a standard college web page can easily be displayed explicitly utilizing the www.college.com hostname.

Despite these successes, several critical technical limitations currently remain. First, the provided visual evidence ultimately represents only a distinctly small simulated topology utilizing precisely one central switch and one core router. Consequently, it completely fails to technically demonstrate any form of hardware redundancy. Second, the captured DHCP configuration screen definitively contains highly specific erroneous values. Third, one recorded server-subnet connectivity ping actively shows exactly one single lost data packet occurring within just four attempts.

Fourth, there is an absolute lack of provided visual screenshots demonstrating reliable email delivery, complex ACL rule enforcement, or functional NAT/PAT routing. Fifth, no quantitative metrics regarding network throughput, general delay, packet jitter, or system recovery benchmarks were mathematically measured during this research. Collectively, these specific operational limitations explicitly prevent the researchers from legally making any quantitative academic claims regarding total system performance or proving comprehensive holistic security based on the current artifact.

IX. ADVANCED SECURITY AND ACCESS CONTROL MECHANISMS

While the initial deployment establishes logical separation through VLANs, an enterprise-grade campus network necessitates a robust, multi-layered security architecture. Expanding upon the baseline configuration, several advanced Layer 2 and Layer 3 security mechanisms should be integrated to protect the campus infrastructure against internal and external threat vectors.

A. Layer 2 Security and Spanning Tree Protection

The access layer, where endpoint devices connect, is particularly vulnerable to malicious activities such as MAC address flooding and unauthorized switch connections. Implementing Port Security is a critical first step. By statically defining or dynamically learning a maximum number of allowed MAC addresses per switch port, administrators can effectively prevent Content Addressable Memory (CAM) table overflow attacks.

Furthermore, the Spanning Tree Protocol (STP), responsible for preventing bridging loops, must be secured. Features such as BPDU Guard should be enabled globally on all PortFast-enabled access interfaces. If an unauthorized switch is attached to an access port and attempts to transmit Bridge Protocol Data Units (BPDUs), BPDU Guard will immediately transition the port into an error-disabled state, thereby preserving the spanning-tree topology and preventing potential network hijacking.

B. DHCP Snooping and Dynamic ARP Inspection (DAI)

In an academic environment, students or staff may inadvertently or maliciously connect unauthorized routers to the network, inadvertently acting as rogue DHCP servers. DHCP Snooping acts as a defensive perimeter by explicitly categorizing switch ports as either "trusted" (facing legitimate DHCP servers or uplinks) or "untrusted" (facing end-user devices). DHCP reply packets are systematically dropped if they originate from untrusted ports.

Building upon the DHCP Snooping binding database, Dynamic ARP Inspection (DAI) should be deployed to mitigate Address Resolution Protocol (ARP) spoofing and poisoning attacks. DAI scrutinizes intercepted ARP packets and validates the IP-to-MAC address bindings against the trusted database. Packets with invalid bindings are immediately discarded, ensuring that attackers cannot seamlessly intercept traffic destined for the default gateway or other legitimate endpoints.

C. Granular Access Control Lists (ACLs)

Moving up to Layer 3, inter-VLAN routing inherently permits all traffic to flow between subnets unless explicitly restricted. Standard and Extended Access Control Lists (ACLs) are strictly required to enforce the principle of least privilege. Extended ACLs offer granular control, allowing filtering based on source IP, destination IP, protocol (TCP/UDP/ICMP), and specific port numbers.

For example, an Extended ACL should be configured and applied inbound on the router's interface servicing the student VLANs. This ACL could explicitly permit HTTP (port 80) and HTTPS (port 443) traffic destined for the server VLAN (192.168.70.0/24), while comprehensively dropping any SSH (port 22) or Telnet (port 23) traffic aimed at the server or administrative networks. All ACL implementations must conclude with an implicit deny rule and be actively validated utilizing comprehensive ping and application-level testing within the simulator.

X. HIGH AVAILABILITY AND QUALITY OF SERVICE (QOS)

The observed network topology, while functional for fundamental connectivity demonstrations, represents a flat topology heavily reliant on single points of failure. In a physical production campus, the failure of the central aggregation switch or the singular Cisco 2911 router would result in catastrophic network-wide downtime. To transition this prototype into a resilient architecture, High Availability (HA) and Quality of Service (QoS) mechanisms must be integrated.

A. Redundancy Protocols and Physical Resiliency

The architecture should be aggressively upgraded to feature a collapsed core or a dedicated distribution layer incorporating at least two multilayer switches. To provide seamless default gateway redundancy for the endpoint clients, a First Hop Redundancy Protocol (FHRP) such as the Hot Standby Router Protocol (HSRP) or the Virtual Router Redundancy Protocol (VRRP) must be configured.

Under HSRP, the dual distribution switches share a singular virtual IP address and virtual MAC address. If the primary active routing device suffers a hardware failure or loses upstream connectivity, the standby device instantly assumes the active routing role. This transition occurs transparently to the end-users in VLANs 10, 20, and 30, preventing significant operational disruptions. Furthermore, multiple physical links between core switches should be logically bundled using EtherChannel (LACP or PAgP) to provide expanded bandwidth and instant failover capabilities without relying strictly on slower STP recalculations.

B. Traffic Prioritization and QoS Policies

Modern campus networks are heavily converged, transporting not only standard data but also real-time, latency-sensitive traffic such as Voice over IP (VoIP) telephony and high-definition video conferencing. Without QoS, a massive file transfer could easily saturate uplink bandwidth, leading to dropped voice packets and severely degraded communication quality.

To resolve this, QoS policies must be implemented at the network edge. Traffic should be classified and marked using Differentiated Services Code Point (DSCP) values as close to the source as possible. Voice traffic should be placed into an Expedited Forwarding (EF) strict priority queue to guarantee low latency and jitter, while standard HTTP traffic is relegated to best-effort forwarding.

C. Wireless Integration and Future Scalability

Finally, the current design is strictly limited to wired Ethernet endpoints. To support a highly mobile student body, a scalable wireless infrastructure utilizing a centralized Wireless LAN Controller (WLC) and Lightweight Access Points (LAPs) should be modeled. The WLC facilitates seamless roaming, unified security policies, and dynamic radio resource management using the CAPWAP protocol. Wireless traffic can be mapped to highly distinct VLANs, physically separating authenticated staff and student devices from a heavily restricted, internet-only guest network. These extensive upgrades represent the necessary evolutionary steps required to transform this baseline topology into a highly resilient, enterprise-ready campus architecture.

XI. CONCLUSION AND FUTURE WORK

This academic paper comprehensively documents a small-scale college campus network prototype successfully developed within Cisco Packet Tracer, deliberately featuring segmented departmental VLANs, an isolated server VLAN, a singular Cisco 2911 router, one central aggregation switch, and dedicated DHCP, DNS, HTTP, and targeted email server roles. The collected observed evidence strictly confirms that the specifically displayed network trunk reliably uses standard 802.1Q encapsulation while carrying essential VLAN traffic. Furthermore, it validates that a standard college web page is actively displayed when accessing the simulated domain via the network endpoints.

Crucially, the captured DHCP screenshot explicitly identifies a severely flawed pool configuration that mandatorily requires direct correction or final confirmation prior to the project legitimately claiming reliable, automated address assignments. Extensive future work should aggressively aim to manually complete the varied DHCP scopes, strictly verify end-to-end DNS and email application transactions, deeply document the exact Layer 3 routing methodology chosen, and validate secure SSH management workflows. A potential secondary design phase could purposefully introduce redundant Layer 2 switching and Layer 3 routing to augment the reliability of the baseline topology. Ultimately, the current state of this project is strictly best understood purely as a foundational educational design, constrained strictly by the visual evidence actively collected.

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